# CPUE project 1

Beth Babcock

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```
library(tidyverse)
library(glmmTMB)
library(MuMIn)
library(DHARMa)
theme_set(theme_bw())
```

For this question, we will use the obsdat dataset, but with the swordfish column (SWO) rather than BUM.

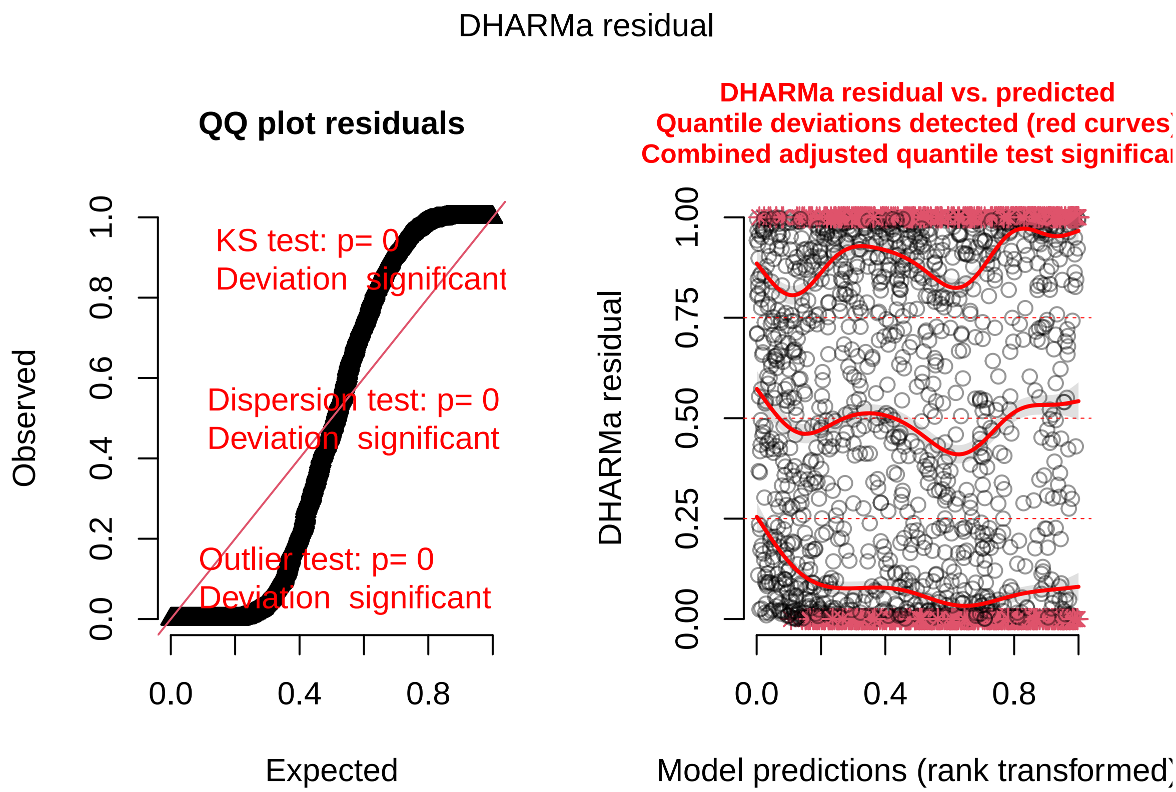
1.Fit a Poisson model with your choice of predictor variables, along with the sets offset, and look at the DHARMa residuals. Does it fit well?

```
obsdat<-read_csv("obsdatSwo.csv")%>%
  mutate(Year=factor(Year),
         season=factor(season),
         area=factor(area))
```

```
## Rows: 1983 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): area
## dbl (9): Year, season, lat, lon, hbf, BUM, SWO, sets, fleet
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
Poisson1<-glmmTMB(SWO~Year+area+season+offset(log(sets)),
                  data=obsdat,
                  family="poisson")
plot(simulateResiduals(Poisson1))
```

```
## DHARMa:testOutliers with type = binomial may have inflated Type I error rates for integer-valued dis
```



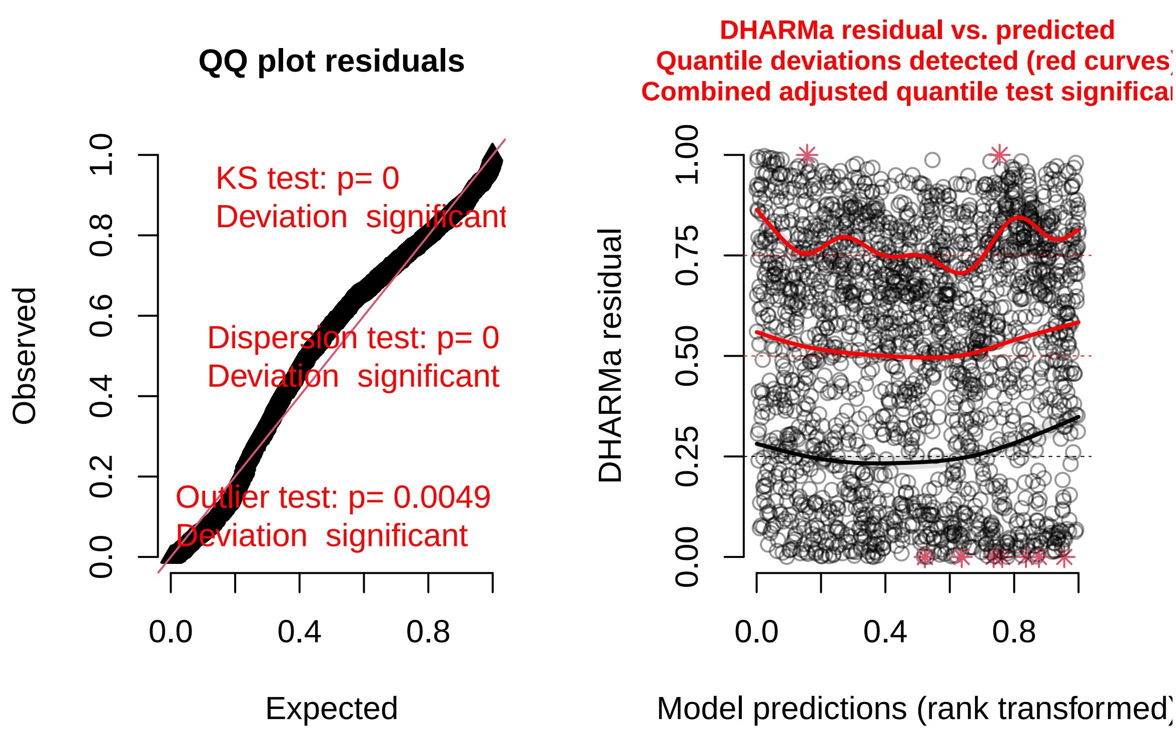

Looks very bad.

2. Fit a negative binomial 2 model and look at the DHARMa residuals. Does it fit well?

```
NB2<-glmmTMB(SWO~Year+area+season+offset(log(sets)),
                  data=obsdat,
                  family="nbinom2")
plot(simulateResiduals(NB2))
```

```
## DHARMa:testOutliers with type = binomial may have inflated Type I error rates for integer-valued dis
```

DHARMa residual




Looks much better.

3. Use the dredge function to find the BIC best model starting the model with all two way interactions. Which model is best?

```
NB2<-glmmTMB(SWO~(Year+area+season)^2+offset(log(sets)),
                   data=obsdat,
                   family="nbinom2",
              na.action="na.fail")
NB2dredge<-dredge(NB2,rank="BIC")
```

```
## Fixed terms are "cond((Int))" and "disp((Int))"
```

```
NB2dredge
```

```
## Global model call: glmmTMB(formula = SWO ~ (Year + area + season)^2 + offset(log(sets)),
##     data = obsdat, family = "nbinom2", na.action = "na.fail",
##     ziformula = ~0, dispformula = ~1)
## ---
## Model selection table
##     cnd((Int)) dsp((Int)) cnd(are) cnd(ssn) cnd(Yer) cnd(are:ssn) cnd(are:Yer)
## 66     0.22210          +        +
## 68     0.27000          +        +        +
## 76     0.25450          +        +        +                     +
## 70     0.33480          +        +                 +
```

```
## 65     0.39550          +
## 72     0.36510          +        +        +        +
## 67     0.41970          +                 +
## 80     0.37600          +        +        +        +            +
## 69     0.51650          +                          +
## 71     0.52410          +                 +        +
## 86    -0.10320          +        +                 +                         +
## 88    -0.07766          +        +        +        +                         +
## 96    -0.06325          +        +        +        +            +            +
## 104    0.12190          +        +        +        +
## 112    0.15290          +        +        +        +            +
## 103    0.31490          +                 +        +
## 120   -0.38180          +        +        +        +                         +
## 128   -0.34410          +        +        +        +            +            +
## 2      3.21600          +        +
## 4      3.25000          +        +        +
## 12     3.27200          +        +        +                     +
## 1      3.39700          +
## 3      3.39800          +                 +
## 6      3.13800          +        +                 +
## 8      3.19300          +        +        +        +
## 16     3.25100          +        +        +        +            +
## 5      3.33700          +                          +
## 7      3.35900          +                 +        +
## 22     2.72300          +        +                 +                         +
## 24     2.76000          +        +        +        +                         +
## 32     2.80200          +        +        +        +            +            +
## 40     2.99000          +        +        +        +
## 48     3.08700          +        +        +        +            +
## 39     3.22000          +                 +        +
## 56     2.51100          +        +        +        +                         +
## 64     2.58900          +        +        +        +            +            +
##     cnd(ssn:Yer) cnd(off(log(sts)))  df     logLik      BIC   delta weight
## 66                                +   3 -7897.020 15816.8    0.00   0.99
## 68                                +   6 -7890.249 15826.1    9.24   0.01
## 76                                +   9 -7884.676 15837.7   20.87   0.00
## 70                                +  31 -7826.915 15889.2   72.38   0.00
## 65                                +   2 -7940.334 15895.9   79.04   0.00
## 72                                +  34 -7819.404 15896.9   80.13   0.00
## 67                                +   5 -7935.163 15908.3   91.47   0.00
## 80                                +  37 -7815.276 15911.5   94.65   0.00
## 69                                +  30 -7859.549 15946.9  130.05   0.00
## 71                                +  33 -7853.003 15956.6  139.74   0.00
## 86                                +  59 -7789.852 16027.7  210.84   0.00
## 88                                +  62 -7783.829 16038.4  221.57   0.00
## 96                                +  65 -7778.086 16049.7  232.86   0.00
## 104            +                  + 118 -7763.699 16423.3  606.48   0.00
## 112            +                  + 121 -7758.453 16435.6  618.77   0.00
## 103            +                  + 117 -7793.092 16474.5  657.68   0.00
## 120            +                  + 146 -7722.318 16553.1  736.30   0.00
## 128            +                  + 149 -7714.794 16560.9  744.03   0.00
## 2                                     3 -8722.416 17467.6 1650.79   0.00
## 4                                     6 -8716.466 17478.5 1661.67   0.00
## 12                                    9 -8711.339 17491.0 1674.19   0.00
```

```
## 1                                         2 -8746.424 17508.0 1691.22   0.00
## 3                                         5 -8739.993 17517.9 1701.13   0.00
## 6                                        31 -8678.466 17592.3 1775.48   0.00
## 8                                        34 -8671.224 17600.6 1783.77   0.00
## 16                                       37 -8665.929 17612.8 1795.96   0.00
## 5                                        30 -8696.988 17621.7 1804.93   0.00
## 7                                        33 -8689.192 17628.9 1812.12   0.00
## 22                                       59 -8642.373 17732.7 1915.88   0.00
## 24                                       62 -8637.419 17745.6 1928.75   0.00
## 32                                       65 -8629.114 17751.7 1934.91   0.00
## 40             +                        118 -8619.824 18135.5 2318.73   0.00
## 48             +                        121 -8611.930 18142.5 2325.72   0.00
## 39             +                        117 -8639.287 18166.9 2350.06   0.00
## 56             +                        146 -8583.721 18275.9 2459.11   0.00
## 64             +                        149 -8573.896 18279.1 2462.24   0.00
## Models ranked by BIC(x)
```

Best model has area only.